VANDERBILT UNIVERSITY

Machine Learning Project Proposal: Using Multi-Class Classification and One-Vs-All Approaches to Processed Interaction Data

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Suyash Deshmukh, Daniel Gormezano, Yi He, Israel Lwamba, Payton Robinette, Jiho Yoo

PROJECT OVERVIEW

For the course project, we plan on using processed data because it allows us to focus on model design and hone performance metrics rather than work on image processing. As seen in **Figure 1**, there is a high imbalance in the class distribution, as drag and click account for around 44.9% and 32.6% of actions, respectively, whereas scrollup, disappear, appear, and scrolldown all account for less than 1% each. Therefore, this imbalance may bias the model towards predicting the majority classes and must be accounted for during training. Furthermore, **Figure 2** reveals distribution-related information about a few features in the dataset. Some variables are clustered around 0 (bbox_area_ratio_delta), while others are on different scales (bbox_aspect_avg). Therefore, tree-based models may have higher performance because they can better handle non-linear data and uneven scales.

Our first approach to this project will be a multi-class classification problem because each observation represents a clear user action label, and incorporating all the actions into one model permits us to learn common patterns across classes while understanding what is different among them. To handle class imbalance, we will use class weights during training. Unlike resampling methods, adding class weights does not introduce synthetic data or remove observations, reducing the risk of overfitting.

We will also implement a one-vs-all approach to train a separate classifier for each action against all others; this will permit each model to distinguish one action at a time, potentially improving the performance for minority classes, such as scrollup and disappear. Since there are 11 action types, we would train 11 one-vs-all models. For instance, one model would predict "click" (class 1) and "not click" (not class 1). For this, we will apply SMOTE to increase the representation of minority classes and create a more balanced training set. These methods will help ensure that the model does not favor classes, as having strong performance on majority classes alone is not a reliable indicator of model quality.

The three models we will use to evaluate model performance on the dataset are Logistic Regression, Random Forest, and XGBoost. Logistic regression provides a baseline model that would help us comprehend how well a linear model performs across outcome labels. We plan on applying logistic regression to the 11 one-vs-all approaches, as the labels are binary outcomes. Random Forest will be used for one of the multi-class classification models because it is robust at capturing non-linear relationships and less affected by unequal feature scaling. Lastly, XGBoost will be applied to the other multi-class classification model, as it is an ensemble boosting method that builds on the performance of weaker learners to obtain stronger performance on imbalanced datasets. To reduce overfitting and improve the generalizability of the model to unseen data, we will perform hyperparameter tuning for Random Forest and XGBoost.

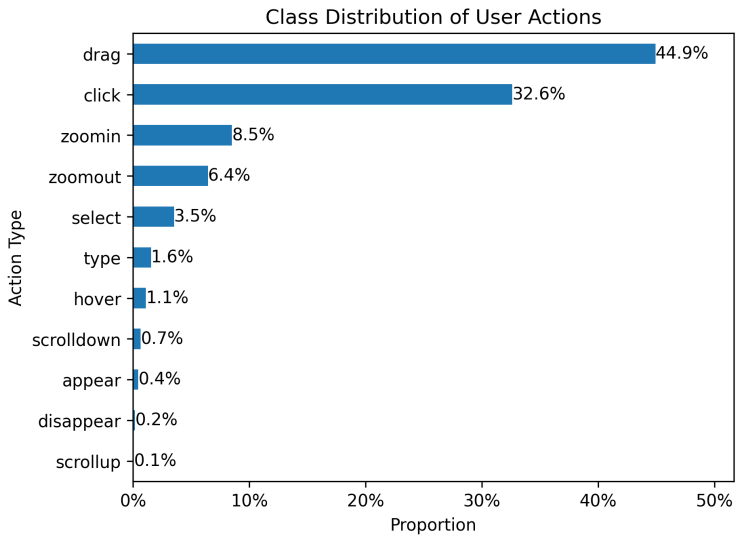

Figure 1: **Class Distribution of User Actions.** This horizontal bar chart shows the percentage proportion of each action type. The x-axis represents the proportion of observations, and the y-axis lists each action type.

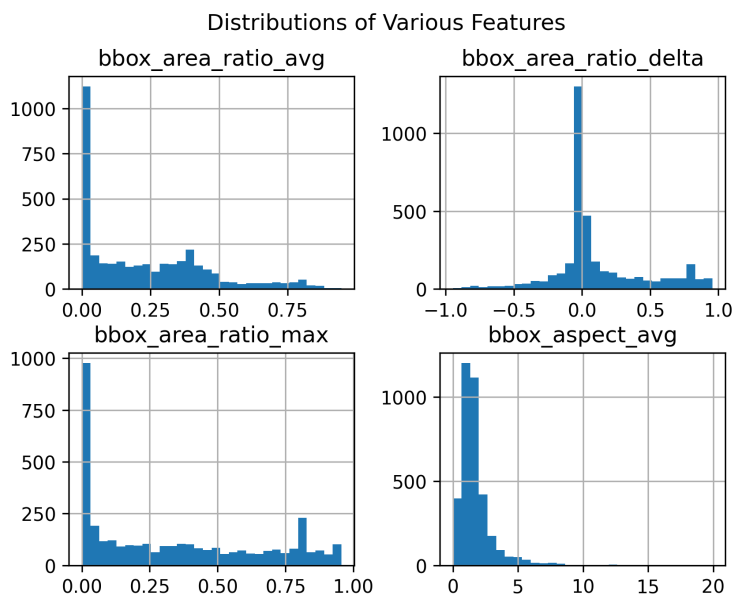

Figure 2: **Histograms of Features in the Processed Dataset.** These faceted histograms reveal the distribution of various features in the dataset. The x-axis represents the range of feature values, and the y-axis shows the count of observations within each bin.